Lab 1

Installation of C++ and Running of the first the program

Learning Objectives

The objective of this lab was to become familiar with programming tools and version control. In this lab we installed a C++ development environment, created and executed basic programs, and learned how to use GitHub to upload our lab work. We also created a tutorial video explaining the steps

Software’s Used

- **Dev C++ / VS Code** – for writing and running C++ programs.
- **GitHub** – for storing and sharing the lab repository online.
- 3. MinGW-w64 (Recommended Updated Version)

- MinGW-w64 https://www.mingw-w64.org/
- https://www.mingw-w64.org/downloads/

- . C/C++ Extension for VS Code

1: Installing Dev C++ / VS Code and Running Basic Programs

First, we installed **Dev C++ / Visual Studio Code** on the computer.

Steps:

1. Downloaded the software from the official website.
2. Installed it on the computer.
3. Opened the IDE (Integrated Development Environment).
4. Created a new C++ file.
5. Wrote a simple C++ program.

2: Creating a GitHub Repository

We created a GitHub account and uploaded the lab files.

Steps:

1. Opened **github.com**.
2. Created a new account.
3. Clicked **New Repository**.
4. Named the repository **Programming-Lab**.
5. Uploaded the **Lab-01 files**.
6. Saved the repository online.

GitHub helps in storing code and sharing it with others.

In this lab we learned how to install a programming environment, write and execute a basic C++ program, and upload our work to GitHub. We also practiced explaining the process by creating a tutorial video. This lab helped us understand the basic tools required for programming.

Lab 1: Tasks

Task 1: Declare Different Types of Variables & Find Their Memory Size, Location, & Values

```
#include <iostream>   // Tells the compiler to include in/output stream library
using namespace std;     // Allows us to use cout and cin without writing std::
int main() {                                    // Main function where program begins
   cout << "This is Safi Ullah!\”;// Display output on screen
              return 0;                         // Indicates program ended successfully
          }
```